Young Children's Understanding of Posterior Probabilities

by

Yilun (Ellen) Tong

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Abstract

The ability to reason about posterior probabilities — updating an initial probabilistic judgment in light of new evidence — is fundamental to both everyday reasoning and scientific thinking. While crucial, little research has investigated the developmental origins of children's ability to reason about posterior probabilities, with one study suggesting that it does not develop until age five (Girotto & Gonzalez, 2008). I investigate young children's ability to reason about posterior probabilities using a new method. Across two experiments, 4- to 6-year-old children (n = 72) played a game where they had to guess the color of a block that a robot sampled from an opaque jar. Children made an initial guess about the blocks' color, followed by a second guess after being told new information regarding the block's shape (posterior probability judgment). I found that, contrary to previous research, children as young as 4 succeed at reasoning about posterior probabilities: they can make accurate probabilistic judgments and revise them based on new evidence. This suggests that the ability to reason about posterior probabilities may emerge earlier than previously thought. Limitations of the current studies and future directions are discussed. To establish a clearer developmental trajectory of children's ability to reason about posterior probabilities, future research may examine whether even younger children can do so.

Keywords: probability updating, posterior probability, decision-making, cognitive development

Introduction

Imagine waking up to a clear, sunny sky. You might initially judge the likelihood of rain to be low and decide against bringing an umbrella. However, upon stepping outside, you notice dark clouds forming. This new evidence prompts you to revise your belief, increasing your estimate of the likelihood of rain. As a result, you adjust your decision and take an umbrella with you.

This everyday scenario demonstrates reasoning about posterior probability. This process involves first making an initial probabilistic judgment (prior probability) and then updating that judgment when presented with new information (posterior probability). In the example above, you first determined a prior probability by concluding there was a low chance of rain. Then, based on the evidence of the dark clouds, you updated your prior probability to a posterior probability, concluding that rain was more likely, and you should therefore bring an umbrella.

The ability to reason about posterior probabilities — to update probabilistic judgments based on new evidence — is fundamental to human reasoning. This skill is essential not only for everyday decision-making but also for critical thinking in fields such as science and law, where adjusting reasoning in response to new evidence is both necessary and constant. Given its importance in both practical and intellectual contexts, understanding when and how this ability develops is crucial for examining the foundations of human reasoning. In this thesis, I review the previous literature on infants' and young children's probabilistic reasoning and ability to reason about posterior probabilities. I then introduce my current studies, which investigate the development of the ability to reason about posterior probability in young children.

Young Children's Probabilistic Reasoning

The ability to reason about posterior probabilities is contingent upon one being able to make accurate probabilistic predictions. A large body of research indicates that even very young infants can reason about probabilities to make inferences, and reason about objects and agents (Denison & Xu, 2010, 2014; Denison & Xu, 2019 for review; Kushnir et al., 2010; Ma & Xu, 2010; Téglás et al., 2007, 2011; Wellman et al., 2016; Xu & Garcia, 2008). For example, 12-month-old infants can identify the more probable outcome when sampling from a lottery machine containing three yellow blocks and one blue block (Téglás et al., 2007). Similarly, infants as young as 8 months old understand random sampling and expect a randomly drawn sample to reflect the population it was sampled (Xu & Garcia, 2008). These findings suggest that probabilistic reasoning abilities are present from a very young age.

Despite infants' success at probabilistic reasoning tasks, some studies show that preschoolers struggle at making explicit probabilistic judgments, suggesting that the development of probabilistic reasoning might not be linear. For example, preschoolers perform at chance levels or rely on heuristics when asked to make probabilistic judgments, particularly in tasks requiring them to compare proportions or integrate multiple sources of information (Girotto & Gonzalez, 2008; Girotto et al., 2016; Placì et al., 2020). Additionally, young children struggle to reason about proportions in random sampling events (Girotto et al., 2016; Placì et al., 2020). It remains an open question of how robust young children's probabilistic reasoning capabilities are.

Young Children's Understanding of Posterior Probability

While many studies have investigated the development of probabilistic reasoning, little research has investigated the development of children's ability to reason about posterior probabilities. One study found that children do not develop this ability until age 5 (Girotto &

Gonzalez, 2008). In their study, children played a betting game in which they had to guess the owner of a chip drawn from a bag. Children were introduced to two characters (e.g., Mr. Red, the owner of the red chips, and Mr. Green, the owner of the green chips) and a set of chips varying in color and shape (e.g., three red circles, two red squares, one green circle, and two green squares; Figure 1a). All the chips were placed into an opaque bag, and the experimenter drew one chip without revealing its identity. Children made two consecutive bets about the color of the block: a prior probability guess (prior trial), followed by a posterior probability guess (posterior trial).

In the prior trial, children were asked to guess the color of the drawn chip by betting on the owner of that chip without knowing its shape. To make an informed guess, children should consider the overall color distribution and bet on the character who owns the more probable color. For example, if the set contained more red chips than green chips, children should bet on Mr. Red (Figure 1a).

Following the prior trial, children made a posterior probability guess (posterior trial) after being told the chip's shape (e.g., the shape was revealed to be a circle). There were two types of posterior trials: posterior confirmation and posterior disconfirmation. Half the children received the posterior confirmation trial, and the other half received the posterior disconfirmation trial. In the posterior confirmation trials, the newly revealed shape confirmed children's prior probability guess. For instance, if children had initially bet on Mr. Red, and the revealed shape did not change which color was more probable, they should continue betting on Mr. Red (Figure 1a). In contrast, the posterior disconfirmation trials used a different chip distribution in which the revealed shape contradicted children's prior probability guess. For example, if children had initially bet on Mr. Red, but the revealed shape (e.g., a circle) indicated that most circles were green, they should now switch their bet to Mr. Green (Figure 1b).

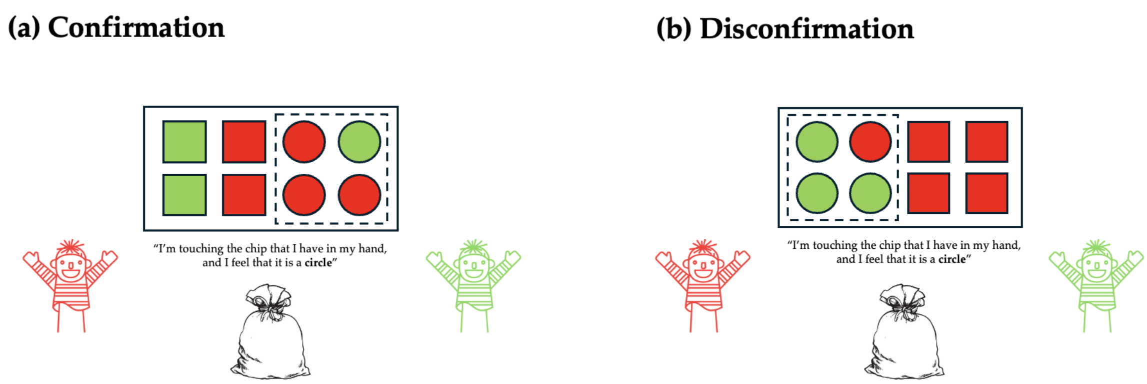

Figure 1. **Girotto & Gonzalez (2008) updating task.** Children first completed the prior trial in which they bet on the owner of the more probable colored chip, and then completed a posterior trial after the shape of the chip was revealed. (a) In the confirmation version, the shape of the block confirmed children's prior probability guesses. (b) In the disconfirmation version, the shape of the block contradicted children's prior probability guesses.

Girotto and Gonzalez (2008) found that all but the youngest children (4-year-old preschoolers) succeeded in the prior and posterior tasks, concluding that children can reason about posterior probabilities by the age of five.

While Girotto & Gonzalez suggest that it is not until age five that children can reason about posterior probabilities and integrate new evidence into their decision-making, there were some important limitations of the study. First, the ratio of chips may have been too small for the younger children to accurately compute the prior probability and make a prior probability guess, leading them to fail the task. In their study, Girotto and Gonzalez (2008) presented children with an array of chips with a color ratio of 5:3 in favor of black in the prior task. It is possible that the 5:3 ratio in their study was not salient enough for children to clearly distinguish between colors and make a probabilistic judgment, as most studies investigating probabilistic reasoning used larger ratios such as 3:1 (Téglás et al. 2007; Xu & Garcia, 2008; Kushnir et al. 2010; etc.). Additionally, research on infants' and young children's probabilistic reasoning suggests that children's ratio discrimination improves with age, and that young children struggle to discriminate small ratios (Libertus & Brannon, 2010; Lipton & Spelke, 2003; Xu & Spelke, 2000; Xu et al., 2005). It is possible that a 5:3 ratio in the prior trials was too small for 4-year-olds to accurately compute the probabilities. Therefore, increasing the ratios in the prior trials may improve 4-year-olds' performance on the task.

Additionally, the use of characters and a betting framework may have added difficulty to the study. In the study, children had to determine which character the randomly drawn chip belonged to, introducing an intermediary agent and an extra layer of complexity. This indirect prediction required children to both represent probabilities and apply them to different characters, potentially making the task more challenging for younger participants. Similarly, asking children to choose which puppet they wanted to be to win further emphasized role-playing elements rather than direct probability judgments. Removing the characters and the betting context may make the study easier for children.

Therefore, the small color ratio and the indirect nature of the prediction introduced unnecessary complexity to reasoning about posterior probability and may have contributed to the results observed in Girotto & Gonzalez (2008). This raises the question of whether younger children could successfully reason about posterior probability if these additional layers of difficulty were eliminated from the task.

Current Study

The current studies introduced several refinements to the original study by Girotto and Gonzalez (2008) to evaluate children's ability to reason about posterior probabilities in a more straightforward manner: (1) the task was simplified by removing characters that were previously used to represent colors, allowing children to make direct color-based probabilistic judgments rather than associating colors with specific characters. (2) The ratio of colors in the prior trials was increased from 5:3 to 12:4 (3:1) to enhance children's ability to discriminate between the more and less probable colors. Additionally, in the posterior confirmation trials, the ratio of the remaining blocks was increased from 3:1 to 100% of the same color. This ensured that the new evidence was fully consistent with one outcome and allowed a clearer test of whether children can attend to the new information and integrate it into their decisions in the posterior trials. (3) A within-subject design was implemented, allowing each child to receive both types of posterior trials.

Finally, to assess children's ability to reason about posterior probabilities (i.e., updating probabilities in light of new evidence), I also conducted additional analyses that were not included in Girotto and Gonzalez (2008). Specifically, I analyzed children's performance on the prior and posterior trials separately and examined how often they correctly answered both trial types, as this would reflect their ability to update their probabilistic judgments. Additionally, I compared children's performance on the posterior confirmation and posterior disconfirmation trials separately to determine whether they performed significantly better on one type of posterior trial than the other.

The current studies aim to examine the ability of 4- to 6-year-old U.S. children to reason about posterior probabilities using a design modified from Girotto & Gonzalez (2008). Children played a game where they had to guess the color of a block that a robot named Earl had sampled from a jar. In the game, children made a prior probability guess about the block's color (prior trial), followed by a posterior probability guess after being told the block's shape (posterior trial). Each trial set began with a prior trial, in which children were shown a set of 12 blocks with two different colors (e.g., eight blue stars, one blue square, three black squares; Figure 3). Next, Earl drew one block, but children couldn't see what the block was. Children were then asked to guess which color the randomly drawn block would be. This trial assessed their ability to make probabilistic judgments based solely on the color distribution.

Following the prior trial, children completed either a posterior confirmation trial or a posterior disconfirmation trial, in which they were told the shape of the block. In the posterior confirmation trials, the shape was associated with the more probable color from the prior trial, confirming children's prior probability guesses. In the posterior disconfirmation trials, the shape was associated with a different color than the one that was more probable in the prior trial, contradicting children's prior probability guesses. I hypothesize that if children can reason about posterior probabilities, they should be able to make an accurate prior probability guess and then update it accordingly when presented with new evidence.

Experiment 1

Method

Experiment 1 was approved by the UC Berkeley Institutional Review Board, and was pre-registered at as.predicted (https://aspredicted.org/2mzn-y8mr.pdf).

Participants

Participants were 51 children aged 5-6 who had a proficient English language understanding. Informed consent for child participation was obtained from parents or legal guardians through the completion of a parental consent form. After the session, each child was awarded a prize (e.g., stickers). Among the 51 participants, one was excluded for failing to identify and name colors, and two were excluded for failing to identify and name shapes before the training trials. All participants were recruited at the Lawrence Hall of Science in Berkeley, California. The final sample included data from 48 children (age in years: M = 6.01, SD = 0.52, range = [5.07, 6.97], 43.75% female).

Materials

The study was conducted using a laptop, with stimuli created through Microsoft PowerPoint animations. Stimuli included animated images of a wooden box containing blocks of various shapes and colors (e.g., blue star, black square), an opaque jar, and a cartoon robot. All data were analyzed and visualized using packages in R (e.g., tidyverse, ggplot2, rstatix, lme4, dplyr).

Design

Experiment 1 adopted a within-subjects repeated measures design. Each child completed four pairs of test trials, resulting in eight test trials in total. In each pair, children first completed a prior trial, where they made a prior probability guess about the color of a specific block from an array of blocks. They then completed a posterior trial, where they made a posterior probability guess after receiving additional information about the block they were guessing. There were two types of posterior trials: (1) a posterior confirmation trial, in which the more probable color in the posterior trial matched the more probable color in the prior trial, and (2) a posterior disconfirmation trial, in which the more probable color in the posterior trial differed from the more probable color in the prior trial.

Children were randomly assigned to one of four counterbalanced orders (Orders A–D). The counterbalancing accounted for two factors: (1) the direction in which blocks were presented and animated for the children (left-to-right vs. right-to-left), and (2) the order in which the two types of trial pairs were presented (confirmation, disconfirmation, disconfirmation, confirmation or disconfirmation, confirmation, confirmation, disconfirmation).

Procedure

Children sat in front of the computer screen, next to the experimenter, for the entire duration of the study. Before the study began, children were asked to name different colors and shapes. Children were excluded from the final analysis if they could not identify colors or shapes.

The study consisted of three parts: two shape training trials, two color training trials, and eight test trials.

Shape Training Trials

Children completed two shape training trials. The purpose of the trials was to ensure that children could identify the dominant shape in the box. In the first trial, children saw a wooden box containing three pink circles and one blue triangle (Figure 1). The experimenter introduced children to the blocks: "Look, here we have a box with some pink blocks and blue blocks. Some of the blocks are circles and some are triangles." As the experimenter introduced the blocks, the corresponding block shapes flashed (i.e., a flashing animation) to draw the child's attention to their shape. Then, children were asked, "Now are there more circles or triangles in the box?" Children responded verbally. If a child answered correctly, the experimenter affirmed their answer by saying, "That's right, there are more circles than triangles." If a child gave an incorrect answer, the experimenter corrected them by saying, "Not quite. See, there are more circles than triangles."

The second shape training trial was the same as the first, except this time there were more triangles in the box. Children were again asked, "Are there more circles or triangles in the box?" The experimenter would then affirm or correct children once they gave an answer. Children who failed both shape training trials were excluded from the study.

Color Training Trials

The procedure for the color training trials was identical to that of the shape training trials, except that children were asked to identify which color was more prominent in the box. In each of the two color training trials, children were shown a box containing red squares and yellow stars (Figure 2) and asked, "Are there more red blocks or yellow blocks in the box?" Children responded verbally. Children who failed both color training trials were excluded from the study.

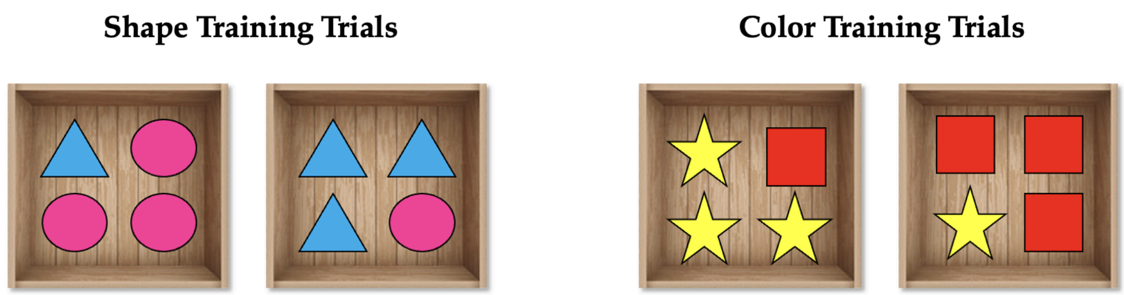

Figure 2. **Shape and color training trials.** Children completed two shape training trials and two color training trials to ensure they could identify the predominant shape or color in a block array.

Upon completing the color training trials, children were awarded three gold stars displayed on the slideshow, and upon completing the test trials, they received eight gold stars regardless of response accuracy.

Children were then told they were about to play the final game. They were shown a transition slide that showed a jar of blocks with various shapes and colors on the left and Earl the Robot on the right, and were told that their job was to figure out the colors of different blocks that Earl drew from the jar. They were also told that the more questions they answered correctly, the more gold stars (displayed at the end of the game) they could get. The experimenter then proceeded to the test trials.

Test Trials

Each child completed eight test trials, divided into four pairs of prior trials and posterior trials. The color and shape of the blocks varied across the trial pairs.

Prior Trial. In each trial pair, children first made a prior probability guess about the color of the block that Earl the Robot sampled from an opaque jar. The purpose of the prior trial was to assess whether children could predict the color of a block by considering the proportion of different colored blocks. At the beginning of each prior trial, children saw Earl the Robot positioned in the top right corner of the screen next to an opaque jar. Below Earl and the jar, a wooden box containing 12 blocks was displayed at the bottom of the slide (e.g., one blue square, three black squares, and eight blue stars; see Figure 3a). The experimenter introduced Earl and the blocks by saying, "Look, here is Earl the Robot, and here we have a box with some [blue] blocks and [black] blocks. Some of the blocks are [stars], and some are [squares]." To draw the child's attention, the corresponding block's shape and color flashed on the screen as the experimenter named each color and shape.

Next, the children watched an animation showing all the blocks moving from the wooden box into the opaque jar. Once the blocks were inside the jar, the box disappeared, and a card appeared displaying images of all the blocks that were placed in the jar. The experimenter explained that the card was meant to help the children remember which blocks were in the jar. The experimenter then clicked on the screen to animate the jar shaking (i.e., mixing the blocks in the jar).

Once the blocks were mixed, Earl approached the jar and grabbed a block, but did not take it out of the jar. The experimenter then asked, "Looking at all the blocks copied on this card here [the experimenter points to the card], do you think Earl's block is [blue] or [black]?" (Figure

3b). The children responded by naming one of the two colors. No feedback was given other than a neutral "Thank you!". The experimenter then proceeded to the posterior trial.

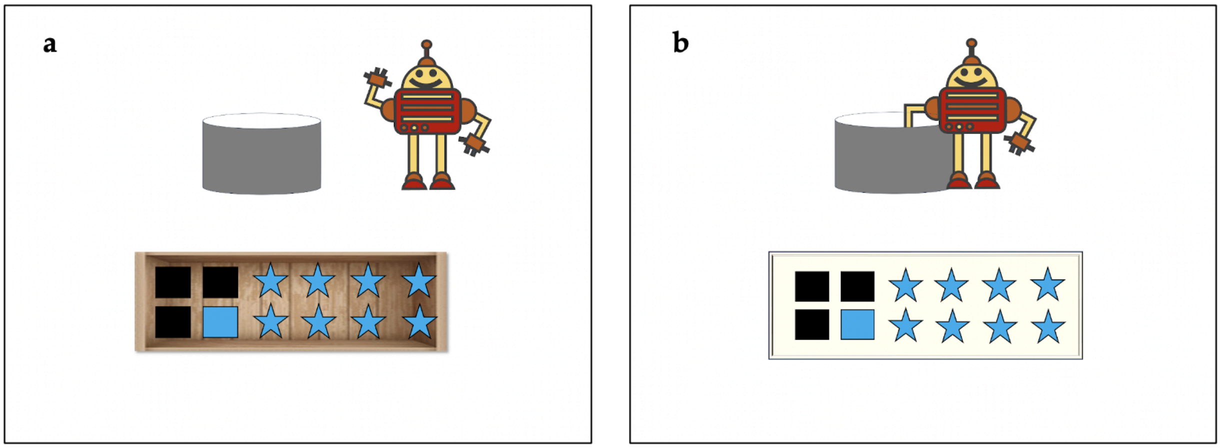

Figure 3. **Test trials.** (a) Children were introduced to Earl the Robot and the blocks. (b) The blocks were placed into the jar but were also copied on a card for reference. Children were asked to guess the color of the block Earl was holding in his hand.

Posterior Trial. After each prior trial, children completed a posterior trial (either confirmation or disconfirmation) where they made a posterior probability guess about the color of Earl's block after the experimenter revealed its shape. The experimenter said to the child, "Listen closely, Earl is going to give us a hint about the block he is holding. Earl felt the block, and he says the block is a [star in a posterior confirmation trial, or square in a posterior disconfirmation trial]. Now that we know Earl's block is a [star / square], do you think Earl's block is [blue] or [black]?" Children named one of the two colors and received no feedback except a "Thank you!".

Half of the posterior trials were posterior confirmation trials, and the other half were posterior disconfirmation trials. In the posterior confirmation trial, the shape of the block corresponded with the more probable color in the prior trials (e.g., the shape of the block Earl was holding was a star, so children should continue to guess blue; see Figure 4a). In the posterior disconfirmation trial, the shape of the block contradicted children's initial prediction (e.g., the shape of the block Earl was holding was a square, so children should change their answer from blue to black; see Figure 4b).

At the conclusion of the study, children were shown eight gold stars and were told they did a great job.

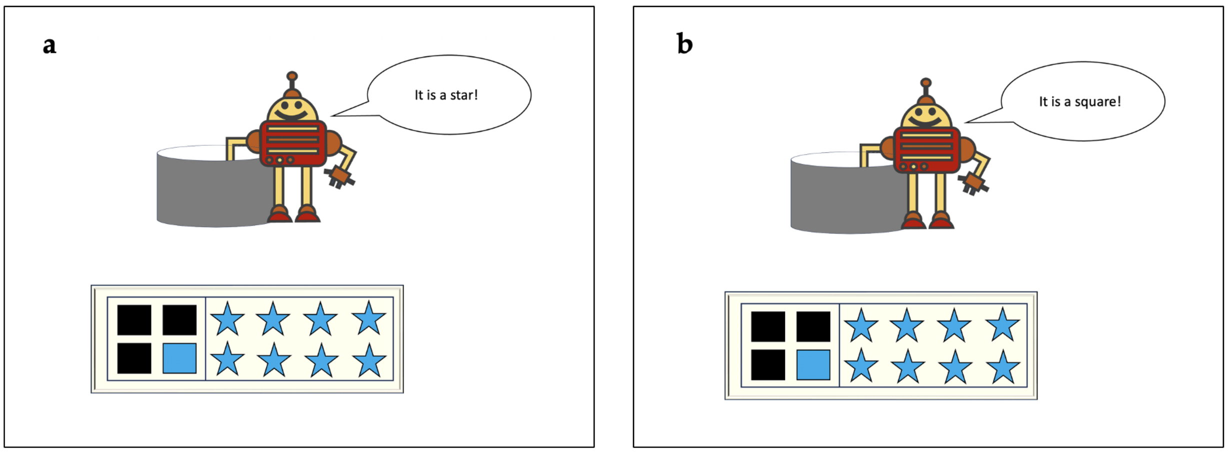

Figure 4. **A posterior confirmation vs. posterior disconfirmation trial.** (a) Assuming a child correctly guessed the block would be blue in the prior trial, the hint that the block is a star will confirm their prior probability guess. (b) The hint that the block is a square will contradict the child's prior probability guess.

Results

My analysis consisted of three parts. First, I conducted a preliminary analysis to test for the effects of age, gender (Male vs. Female), order (A-D, as described in the design section), and trial number (1-8) on children's performances in the test trials. Second, I compared children's performances on the three types of test trials (prior, posterior confirmation, and posterior disconfirmation) to chance (50%). Finally, I examined how often children correctly updated probabilities – that is, how often they provided correct responses for the prior probability guess and subsequently maintained or revised their guess accurately during the posterior trial based on the evidence presented. To do so, I first examined how often children got both the prior and posterior trials correct. I then analyzed posterior trial performance in children who answered the majority of prior trials (i.e., at least three out of four) correctly, ensuring the sample included children capable of probabilistic reasoning.

Preliminary Analysis

A Generalized Linear Mixed Effects Model (GLMM) was fitted, predicting the probability of a correct response from the fixed effects of age, gender, order, and trial number, with participant ID as the random intercept. There was no significant main effect of age ($\widehat{\beta}$ = 0.09, SE = 0.26, z = 0.33, p = 0.74), gender ($\widehat{\beta}$ = 0.39, SE = 0.26, z = 1.47, p = 0.14), order (Order B: $\widehat{\beta}$ = -0.30, SE = 0.40, z = -0.76, p = 0.45; Order C: $\widehat{\beta}$ = -0.40, SE = 0.38, z = -1.04, p = 0.30; Order D: $\widehat{\beta}$ = -0.36, SE = 0.38, z = -0.95, p = 0.34), or trial number ($\widehat{\beta}$ = 0.02, SE = 0.06, z = 0.36, p = 0.72) on children's performance. Thus, age, gender, order, and trial number did not have any significant effect on participants' performances.

Test Trials Analysis

Chance was set at 50% as children had to choose between one of two colors in each trial. Separate Wilcoxon signed-rank tests were conducted to compare children's performance on each trial type (prior, posterior confirmation, and posterior disconfirmation) against the 50% chance level.

The average accuracy for 5- and 6-year-olds (n = 48) on the prior trials was 76% (SD = 0.20), significantly above chance (n = 48, V = 691, p < 0.001, r = 0.81; see Figure 5). The average accuracy for 5-year-olds (n = 25) was 74% (SD = 0.18), significantly above chance (V = 171, p < 0.001, r = 0.82). The average accuracy for 6-year-olds (n = 23) was 77% (SD = 0.21) , which was also significantly above chance (V = 184, p < 0.001, r = 0.80).

The average accuracy for 5- and 6-year-olds on the posterior confirmation trials was 96%, significantly above chance (V = 990, p < 0.001, r = 0.96). The average accuracy for 5-year-olds was 94% (SD = 0.17), significantly above chance (V = 253, p < 0.001, r = 0.94). The average accuracy for 6-year-olds was 87% (SD = 0.10), which was also significantly above chance (V = 253, p < 0.001, r = 0.98).

The average accuracy for 5- and 6-year-olds on the posterior disconfirmation trials was 71% (SD = 0.38), significantly above chance (V = 518, p < 0.001, r = 0.48). The average accuracy for 5-year-olds was 70% (SD = 0.40), significantly above chance (n = 25, V = 157.5, p = 0.01, r = 0.45). The average accuracy for 6-year-olds was 72% (SD = 0.36), which was also significantly above chance (V = 110.5, p = 0.01, r = 0.52).

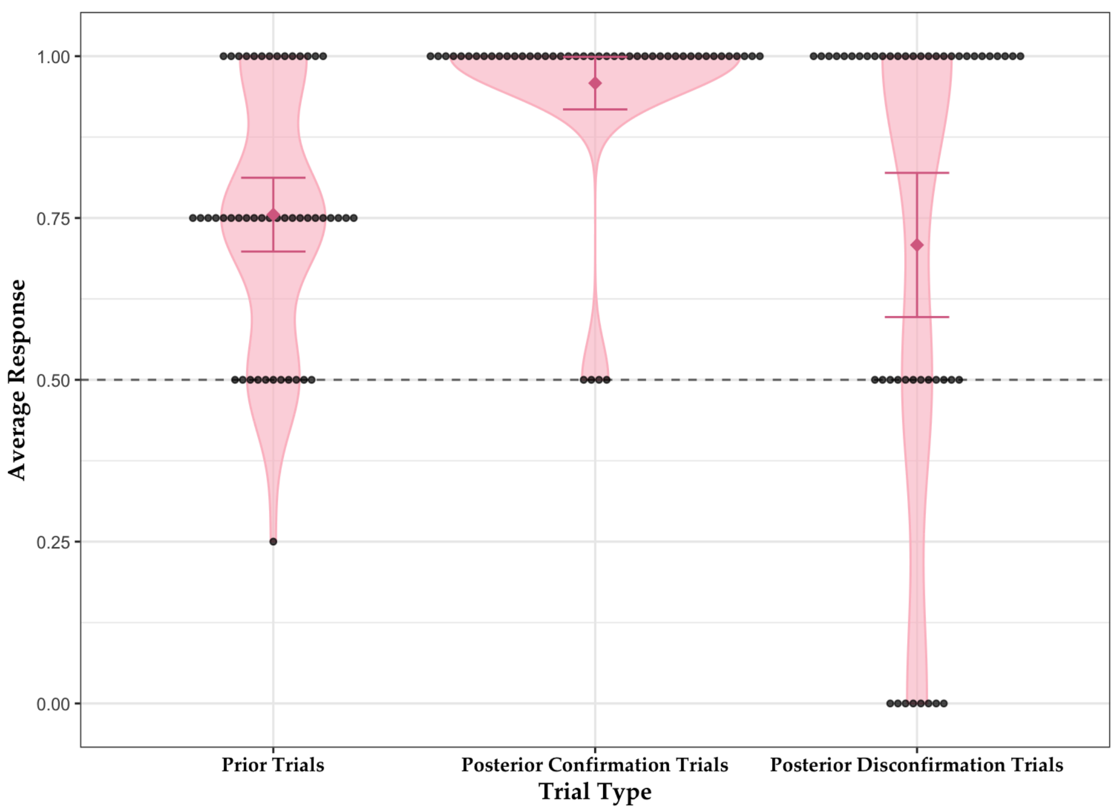

Figure 5. **5- and 6-year-olds' performances in each type of trial.** The black dots represent each child's average response accuracy; the dashed line represents chance level (50%), and the pink violin plots show their distribution along with a 95% confidence interval. Children performed above chance in all three types of trials.

Probability Updating Analysis

The probability updating analysis analyzed how often children got the prior and posterior trials correct in a given trial pair to examine whether they could update probabilities using new information provided. Chance was established at 25% as there were four possible response patterns in a trial pair: (1) the child answered both the prior and posterior trials correctly, (2) the child answered the prior trial correctly but the posterior trial incorrectly, (3) the child answered the prior trial incorrectly but the posterior trial correctly, or (4) the child answered both the prior and posterior trials incorrectly.

Wilcoxon signed-rank tests were conducted to compare children's probability updating performance on the trial pairs against chance. Among the 5- and 6-year-olds, the average accuracy for getting both the prior and posterior trial correct was 61% (SD = 0.22), significantly above chance (V = 978.5, p < 0.001, r = 0.86).

Looking at trial pairs that included a prior trial and a posterior confirmation trial, children's average accuracy was 70% (SD = 0.46), significantly above chance (V = 1135.5, p < 0.001, r = 0.84; see Figure 6). Looking at trial pairs that included a prior trial and a posterior disconfirmation trial, children's average accuracy was 53% (SD = 0.50), which was also significantly above chance (V = 983.5, p < 0.001, r = 0.61). These results suggest that children successfully updated probabilities in both the posterior confirmation and disconfirmation trials.

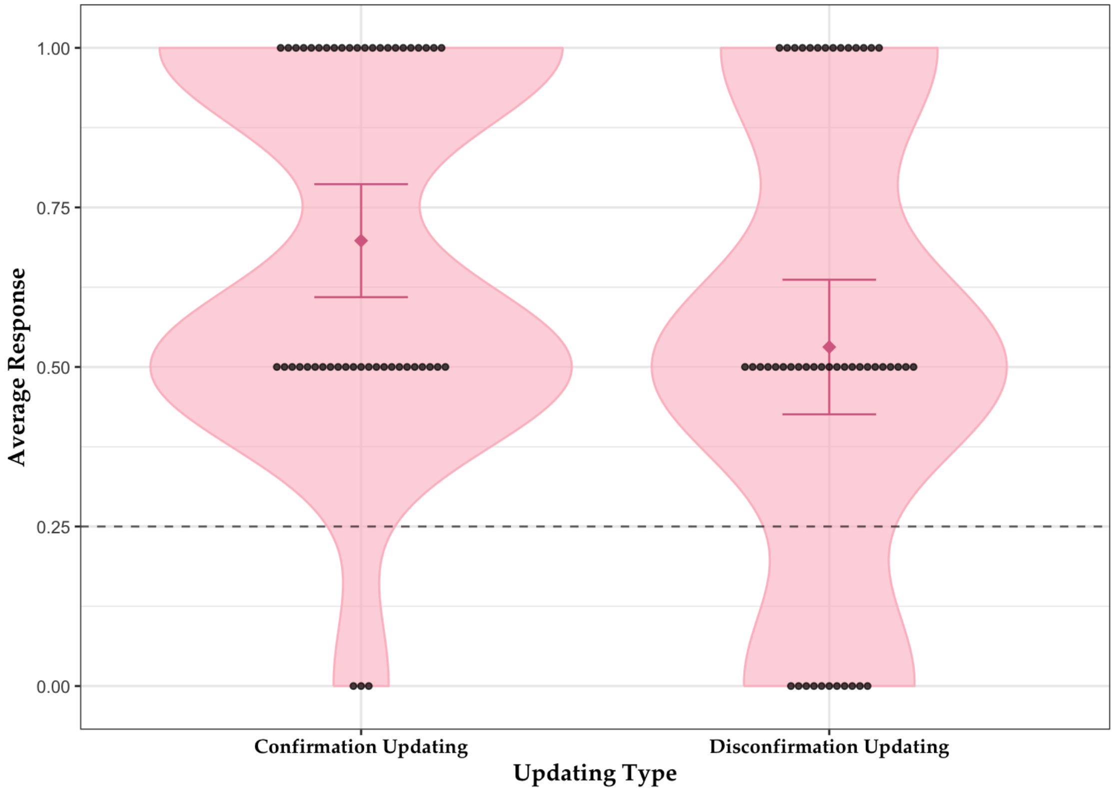

Figure 6. **Probability updating accuracy among 5- and 6-year-olds.** This graph shows children's performance on trial pairs that included a posterior confirmation trial (confirmation updating) and those that included a posterior disconfirmation trial (disconfirmation updating). Children performed above chance in both conditions.

Finally, I wanted to examine only the children who succeeded robustly in making prior probability guesses. Therefore, a filter criterion was applied based on their performance on prior trials. Children who correctly answered at least three out of four prior trials (i.e., accuracy ≥ 75%) were included in the analysis. The accuracy of these children on each trial type was then compared against chance (50%).

Of the 48 participants, 36 met this criterion. For these participants, the average accuracy was 85% (SD = 0.12) on the prior trials, significantly above chance ($V = 666$, $p < 0.001$, $r = 0.90$). Looking only at the 5-year-olds ($n = 18$), the average accuracy on the prior trials was 83% (SD = 0.12), significantly above chance ($V = 171$, $p < 0.001$, $r = 0.91$). Looking only at the

6-year-olds (n = 18), the average accuracy was 86% (SD = 0.13), which was also significantly above chance (V = 171, p < 0.001, r = 0.91).

In the posterior confirmation trials, the average accuracy was 97% (SD = 0.12), significantly above chance (V = 595, p < 0.001, r = 0.97). Looking only at the 5-year-olds, the average accuracy on the posterior confirmation trials was 94% (SD = 0.16), significantly above chance (V = 136, p < 0.001, r = 0.94). Looking only at the 6-year-olds, the average accuracy was 100% (SD = 0), which was also significantly above chance (V = 171, p < 0.001, r = 1).

Finally, in the posterior disconfirmation trials, the 36 5- and 6-year-olds scored an average accuracy of 68% (SD = 0.40), significantly above chance (V = 280, p = 0.006, r = 0.42). Looking only at the 5-year-olds, the average accuracy on the posterior disconfirmation trials was 67% (SD = 0.42), marginally above chance (V = 75, p = 0.06, r = 0.38). Looking only at the 6-year-olds, the average accuracy was 69% (SD = 0.39), significantly above chance (V = 70, p = 0.03, r = 0.46).

These results suggest that 5- and 6-year-olds who consistently performed well in the prior trials also performed well in the posterior trials, demonstrating their ability to effectively update probabilities when presented with both confirming and contradicting information.

Discussion

Results from Experiment 1 indicate that 5- and 6-year-olds can reason about posterior probabilities and update their initial probability judgments in light of new evidence. In the prior trials, children accurately identified the more probable color of Earl's block, demonstrating foundational probabilistic reasoning skills. In the posterior trials, they successfully adjusted their guesses, either confirming their initial inference when the evidence (i.e., the shape of the block) was consistent or revising it when the evidence was contradictory. These findings suggest developmental competence in probability updating.

Given the robust success of the 5- and 6-year-olds on the task, I wanted to see if even younger children were capable of reasoning about posterior probabilities. Therefore, in Experiment 2, I tested whether this ability was present in 4-year-olds.

Experiment 2

Method

Experiment 2 was approved by the UC Berkeley Institutional Review Board, and was pre-registered at as.predicted (https://aspredicted.org/gd9q-833k.pdf).

Participants

Participants were 29 4-year-old children proficient in English understanding. Informed consent for child participation was obtained from parents or legal guardians through the completion of a parental consent form. After the session, each child was awarded a prize (e.g., small race cars, fidget toys, slinkies, and stickers). Among the 29 participants, two were excluded for failing the shape training trials, one for failing the color training trials, one for failing both the shape and color training trials, and another for falling outside the age range for Experiment 2. As in Experiment 1, all participants were recruited at the Lawrence Hall of Science in Berkeley, California. The final sample included data from 24 children (age in years: M = 4.45, SD = 0.27, range = [4.02, 4.99], 75% female).

Materials

The stimuli were identical to those in Experiment 1. Data analysis procedures were consistent across both experiments.

Design

Experiment 2 utilized a within-subject repeated measures design, consistent with Experiment 1. The same four orders were also applied in Experiment 2.

Procedures

The procedures were identical to those in Experiment 1. All children completed the shape training trials, the color training trials, and the eight test trials (organized into four prior-posterior pairs).

Results

The analysis was the same as Experiment 1 and consisted of three parts. First, the effects of age (centered and set as a continuous variable), gender (Male vs. Female), order (A-H as described in the design), and trial number (1-8) were examined to determine whether these factors significantly influenced children's performance on the test trials. Second, children's performance on each trial type was compared against chance. Third, their updating performance was analyzed and compared against chance.

Preliminary Analysis

The effects of age, gender, order, and trial number on children's performance were examined in the preliminary analysis. I examined the main effects of age (in decimals), gender (male vs. female), order (A-D), and trial number (1-8). A Generalized Linear Mixed Effects Model (GLMM) was fitted, predicting the probability of a correct response from the variables described above, with participant ID as the random intercept. The results revealed no significant effects of age ($\widehat{\beta}$ = 0.39, SE = 0.64, z = 0.61, p = 0.54), gender ($\widehat{\beta}$ = 0.34, SE = 0.42, z = 0.81, p = 0.42), order A, B, and D (all p > 0.10), or trial number ($\widehat{\beta}$ = 0.04, SE = 0.07, z = 0.50, p = 0.62) on children's performance. Order C only showed a marginally significant effect on performance (

$\hat{\beta}$ = -1.05, SE = 0.49, z = -2.12, p = 0.03). Age, gender, order, and trial number did not have significant effects on participants' performances.

Test Trials Analysis

Chance was set at 50% because children selected one of two colors in each trial. Wilcoxon signed-rank tests were conducted to compare children's performance on each type of trial against the 50% chance. The average accuracy for 4-year-olds (n = 24) on the prior trials was 71% (SD = 0.27), significantly above chance (V = 95.5, p = 0.003, r = 0.63; see Figure 7). The average accuracy for 4-year-olds on the posterior confirmation trials was 96% (SD = 0.14), significantly above chance (V = 253, p < 0.001, r = 0.96). 4-year-olds' average accuracy on the posterior disconfirmation trials was 50% (SD = 0.42), which was not different from chance (V = 68, p = 0.51, r = 0). 4-year-olds succeeded in the prior and posterior confirmation trials, but not the posterior disconfirmation trials.

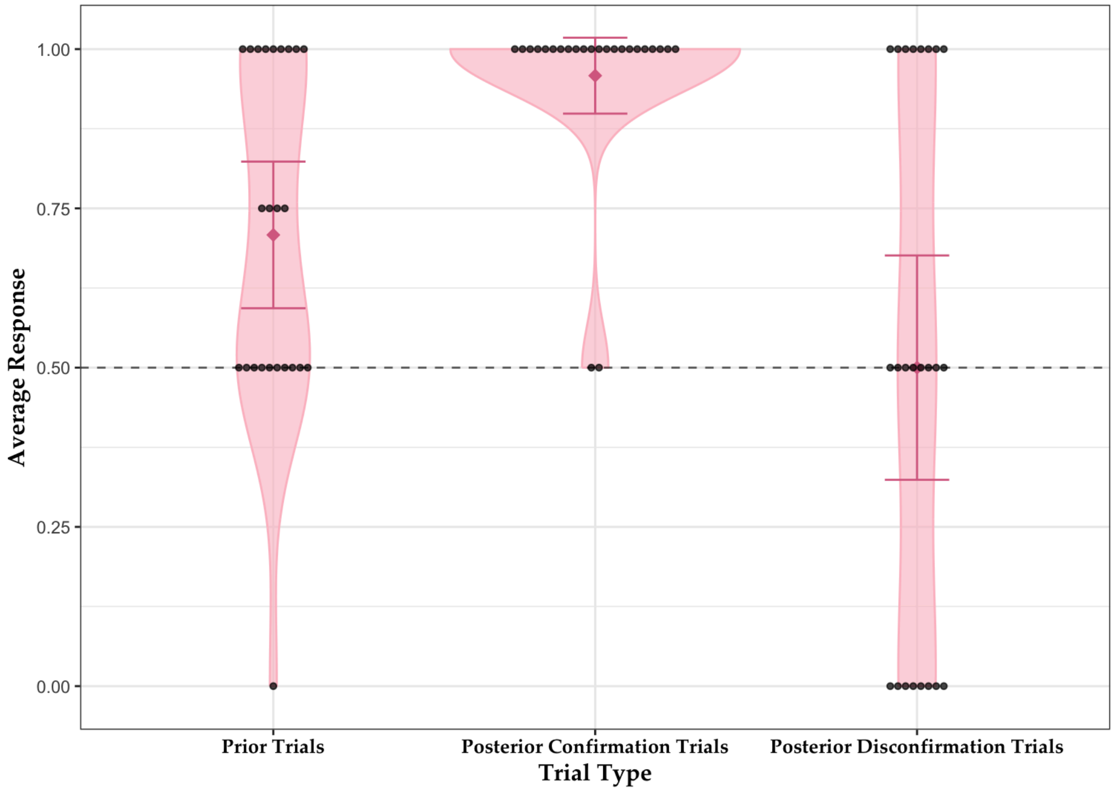

Figure 7. **4-year-olds’ performances in each type of trial.** The black dots represent each child’s average response accuracy; the dashed line represents chance level (50%), and the pink violin plots show their distribution along with a 95% confidence interval. 4-year-olds performed above chance on the prior trials and the posterior confirmation trials, but not the posterior disconfirmation trials.

Probability Updating Analysis

In the probability updating analysis, I examined how often children answered both the prior and posterior trials correctly. As in Experiment 1, chance was established at 25%. Wilcoxon signed-rank tests were conducted to compare children’s performance in the trial pairs against the 25% chance.

Across all the trial pairs, 4-year-olds' average accuracy was 47% (SD = 0.24), significantly above chance (V = 146, p < 0.001, r = 0.73). Looking at trial pairs that included a prior trial and a posterior confirmation trial, children’s average accuracy was 65% (SD = 0.48; see Figure 8), significantly above chance (V = 272, p < 0.001, r = 0.73). Looking at trial pairs that included a prior trial and a posterior disconfirmation trial, children's average accuracy was 29% (SD = 0.46), which was also significantly above chance (V = 157, p = 0.42, r = 0.04). These results suggest that 4-year-olds performed above chance when updating probabilities. However, this was because most children succeeded in the prior and posterior confirmation trials only. The 4-year-olds' disconfirmation updating performance was at chance level.

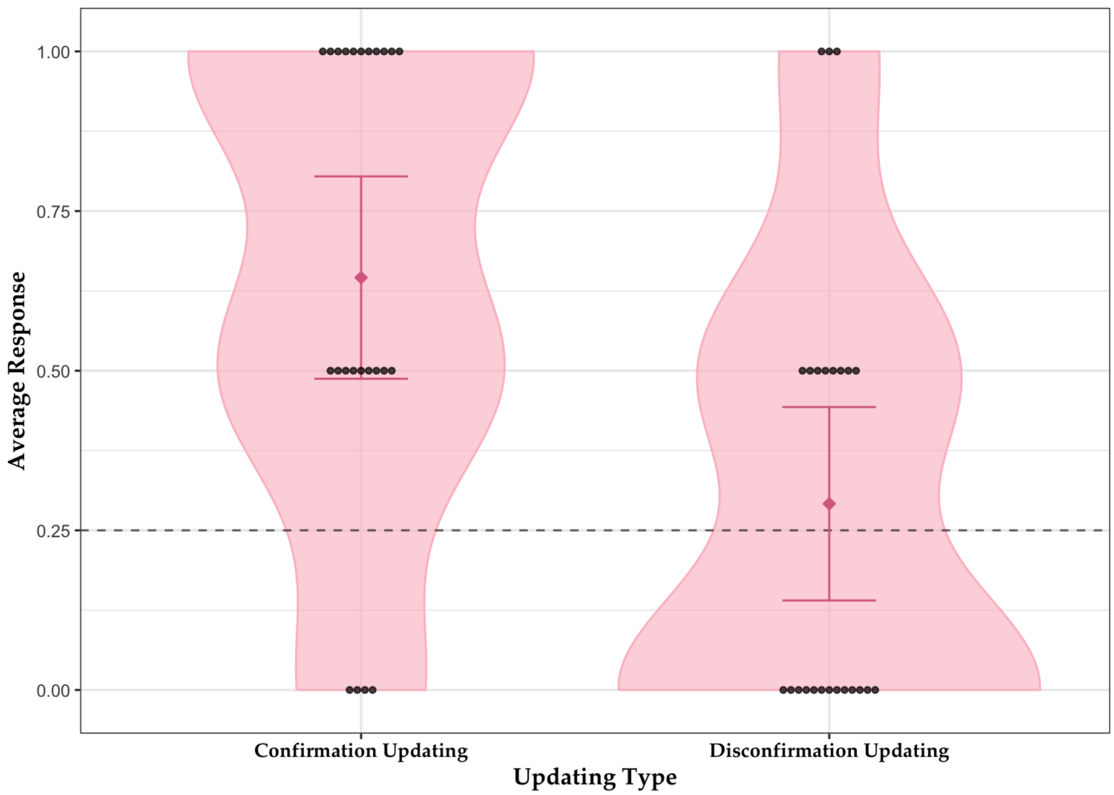

Figure 8. **Probability updating accuracy among 4-year-olds.** This graph shows children's performance on trial pairs that included a posterior confirmation trial (confirmation updating) and those that included a posterior disconfirmation trial (disconfirmation updating). Children performed above chance in the confirmation condition but at chance in the disconfirmation condition.

Consistent with Experiment 1, a filter criterion was applied to their performance on prior trials before analyzing their performance on posterior trials. Only children who answered at least three out of four prior trials correctly (i.e., accuracy ≥ 75%) were included in the analysis. These children's accuracy on each trial type was then compared against chance (50%).

Of the 24 participants, 13 met this criterion. For these participants, the average accuracy was 92% (SD = 0.12) on the prior trials, significantly above chance (V = 91, p < 0.001, r = 0.92). The average accuracy for these 13 4-year-olds on the posterior confirmation trials was 96% (SD = 0.14), significantly above chance (V = 78, p < 0.001, r = 0.96). Finally, their average accuracy on the posterior disconfirmation trials was 35% (SD = 0.43), which was not different from chance (V = 16.5, p = 0.91, r = 0.35). These results suggest that 4-year-olds who consistently performed well on the prior trials also performed well in the posterior confirmation trials, but not in the posterior disconfirmation trials.

Exploratory Analysis

To examine whether children adjusted their predictions in light of new evidence, I conducted an exploratory analysis to compare children's performance on the prior trials, posterior confirmation trials, and posterior disconfirmation trials across the three age groups (i.e., 4-, 5-, and 6-year-olds) from the two experiments (Figure 9). I looked specifically at children's performance on the two types of posterior trials to see if they updated their probabilistic judgments accurately.

Three separate Generalized Linear Mixed Effect Models (GLMM) were fitted to predict children's responses in each trial type with their age (in decimals) as a fixed effect and a random intercept for participant ID. Age did not have a significant effect on children's accuracy in the prior trials ($\widehat{\beta}$ = -0.30, SE = 0.48, z = -0.63, p = 0.53) or in the posterior confirmation trials ($\widehat{\beta}$ = 0.34, SE = 1.52, z = 0.23, p = 0.82), but had a marginally significant effect on children's accuracy on the posterior disconfirmation trials ($\widehat{\beta}$ = -0.42, SE = 1.06, z = -0.39, p = 0.07).

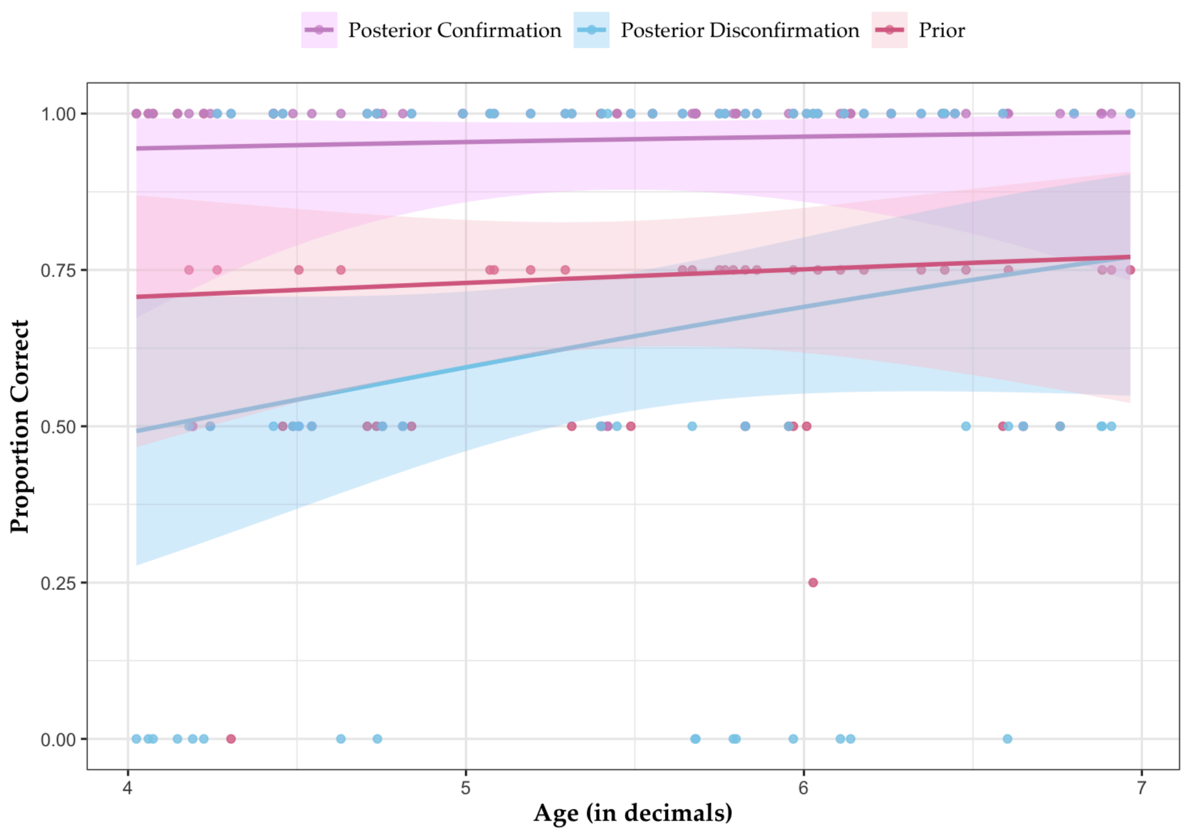

Figure 9. **Average accuracy across the prior, posterior confirmation, and posterior disconfirmation trials for 4-, 5-, and 6-year-olds**. A Generalized Linear Model (GLM) was used to examine age effects. Performance in the posterior disconfirmation trials showed a marginally significant increase with age.

In addition, I wanted to further examine whether children could accurately update their probabilistic judgments when probabilities changed. Therefore, I fitted three separate Generalized Linear Mixed Effect Models (GLMM) to compare children's performance in the posterior confirmation and posterior disconfirmation trials to their performance in the prior trials across the three age groups. The fixed effects in the models were the trial types (prior vs. posterior confirmation or prior vs. posterior disconfirmation).

Comparing children's performance in the posterior confirmation trials to the prior trials, children in all three age groups performed significantly better in the posterior confirmation trials: 4-year-olds ($\widehat{\beta}$ = -2.39, SE = 0.79, z = -3.06, p = 0.002), 5-year-olds ($\widehat{\beta}$ = -1.70, SE = 0.63, z =

-2.68, p = 0.007), and 6-year-olds ($\widehat{\beta}$ = -2.62, SE = 1.04, z = -2.50, p = 0.01). This suggests that children were able to adjust their predictions when probabilities changed and confirmed their previous predictions.

Comparing children's performance in the posterior disconfirmation trials to the prior trials, there was a significant effect of trial type in 4-year-olds ($\widehat{\beta}$ = 8.87, SE = 3.65, z = 2.42, p = 0.01). This suggests 4-year-olds had a hard time adjusting their guesses when probabilities changed and contradicted their previous predictions. There were no significant effects of trial type for 5-year-olds ($\widehat{\beta}$ = 0.19, SE = 0.38, z = 0.52, p = 0.60) or 6-year-olds ($\widehat{\beta}$ = 0.29, SE = 0.41, z = 0.69, p = 0.48).

Discussion

The results from Experiment 2 suggest that four-year-olds successfully guessed the more probable color in the prior trials without any information about the shape of the block, indicating that they could make accurate probabilistic judgments. They also successfully identified the more probable color in the posterior confirmation trials, where the shape corresponded to the more probable color from the prior trials. However, they performed at chance in the posterior disconfirmation trials, where the shape contradicted the more probable color from the prior trials.

When examining probability updating performance, 4-year-olds were able to update their predictions in the posterior confirmation trials but not in the posterior disconfirmation trials. Only about half of the 4-year-olds consistently succeeded in the prior trials (i.e., answering at least three out of four trials correctly). Among these children, successful probability updating was observed only in the posterior confirmation trials.

These findings suggest that while 4-year-olds could update their probabilistic judgments when presented with confirming evidence, they struggled to update when confronted with contradictory evidence.

General Discussion

This study examined children's ability to reason about posterior probabilities, a skill essential for both everyday decision-making and scientific reasoning. Findings from two experiments suggest that 4- to 6-year-olds were able to make accurate probabilistic judgments and update them when new information became available. In Experiment 1, 5- and 6-year-olds made accurate prior probability guesses about the more probable block color and successfully updated them in their posterior probability based on new shape information, whether confirming or contradicting their prior probability guesses. In Experiment 2, 4-year-olds also made accurate prior probability guesses and updated them when the new information confirmed them, though they struggled when the new information contradicted their prior probability guesses. These results suggest that, contrary to previous findings, children as young as four can reason about posterior probabilities.

The success of 5- and 6-year-olds in Experiment 1 replicates findings from Girotto and Gonzalez (2008). Children accurately made a prior probability guess when presented with only the population of blocks and successfully updated it when new evidence either confirmed (posterior confirmation) or contradicted (posterior disconfirmation) their prior probability guess. Notably, the 5- and 6-year-olds in the present study outperformed those in Girotto and Gonzalez (2008), performing significantly above chance on all trial types. This suggests 5- and 6-year-olds have robust ability to reason about posterior probabilities.

The success of 4-year-olds in the prior and posterior confirmation trials contrasts with the findings of Girotto and Gonzalez (2008), in which 4-year-olds performed at chance levels in both the prior and posterior trials. Contrary to Girotto & Gonzalez (2008), 4-year-olds in the current studies performed above chance in the prior and posterior confirmation trials, though not in the posterior disconfirmation trials. Specifically, 4-year-olds in the current studies made accurate prior probability guesses and successfully updated them when presented with information that confirmed their prior probability guesses, though not when the information contradicted their prior probability guesses.

What could account for 4-year-olds' success in the current studies but not in Girotto and Gonzalez (2008)? There are a few possibilities. First, our studies had a larger color ratio in the prior trials. The current studies used a 3:1 ratio (e.g., 12 blue blocks vs. 4 black blocks), whereas Girotto and Gonzalez (2008) used a 5:3 ratio. The larger ratio in the current studies may have made the color distribution more salient, facilitating children's ability to compute probabilities more efficiently. Second, the posterior confirmation trials in the current studies had a probability of 100%, whereas in Girotto and Gonzalez (2008), they had a probability of only 75% (a 3:1 color ratio). The increased probability in the current studies may have made the posterior confirmation trials easier, contributing to children's improved performance. However, raising the probability from 75% to 100% ensured that children were at least attending to the stimuli and integrating the new shape information. Results from exploratory analyses show that children performed significantly better in posterior confirmation trials than in prior trials, indicating that they actively incorporated new evidence into their probabilistic judgments.

While 4-year-olds performed better in the prior and posterior confirmation trials, their failure in the posterior disconfirmation trials replicated the findings from Girotto & Gonzalez

(2008), in which 4-year-olds also performed at chance level in these trials. Although children were able to make accurate prior probability guesses about the color of the block, they struggled when new shape information contradicted their prior probability guesses.

There are at least two possible explanations for children's failure on the posterior disconfirmation trials. First, 4-year-olds may have struggled to switch to a different answer after forming an initial judgment. While children could simply maintain their prior probability guesses in the posterior confirmation trials, the posterior disconfirmation trials require children to revise their prior probability guesses. Children may be reluctant or may lack the cognitive flexibility necessary to update their answer when new information contradicts their original probabilistic judgment. Another possibility is that young children find it challenging to compute probabilities when presented with a small set of objects. In both the current studies and Girotto and Gonzalez (2008), children were shown only four blocks in the posterior disconfirmation trials, with a 3:1 color ratio. If this set size is too small for children to compute probabilities, then increasing the set size should improve their performance on the task.

To adjudicate between these two possible explanations, I am piloting a follow-up study (Experiment 3) to determine whether increasing the set size improves children's performance. If children struggled with probability computations due to the small set size, increasing it should lead to improvement. However, if children were simply having a hard time switching between answers, increasing the set size should have no effect. In this follow-up study, both the posterior confirmation trials and the posterior disconfirmation trials now feature a larger set size of ten objects (within an array of 36 blocks total) with a 4:1 ratio (Figure 10b). This will help determine whether set size is a limiting factor in children's probabilistic reasoning.

As a second modification in the follow-up study, I adjusted the posterior confirmation trials to further investigate younger children's reasoning capabilities under more probabilistic conditions. Since children showed ceiling performance in the 100% probability posterior confirmation trials, suggesting they could attend to the stimuli and integrate new shape information into their probabilistic judgments, I modified both the posterior confirmation trials and the posterior disconfirmation trials to have a 80% probability. This adjustment tests whether children can still succeed when the posterior confirmation trials are probabilistic rather than certain. Separate arrays are now used for the posterior confirmation and posterior disconfirmation trials to ensure equal probabilities in both types of posterior trials. For example, in a posterior confirmation trial, the array includes 19 blue squares, eight blue stars, two black stars, and seven black squares, with the drawn block revealed to be a star (Figure 10a). The color ratio among the stars is 4:1. In a posterior disconfirmation trial, the array consists of two blue squares, 25 blue stars, one black star, and eight black squares, with the drawn block revealed to be a square (Figure 10b). The color ratio among the squares is also 4:1. If children can reason about posterior probabilities, they should be able to make correct prior and posterior predictions even when the trials are probabilistic rather than certain. Results from Experiment 3 would help clarify the factors that hinder children's probabilistic reasoning and determine whether younger children, such as 4-year-olds, can successfully update their probabilistic judgments.

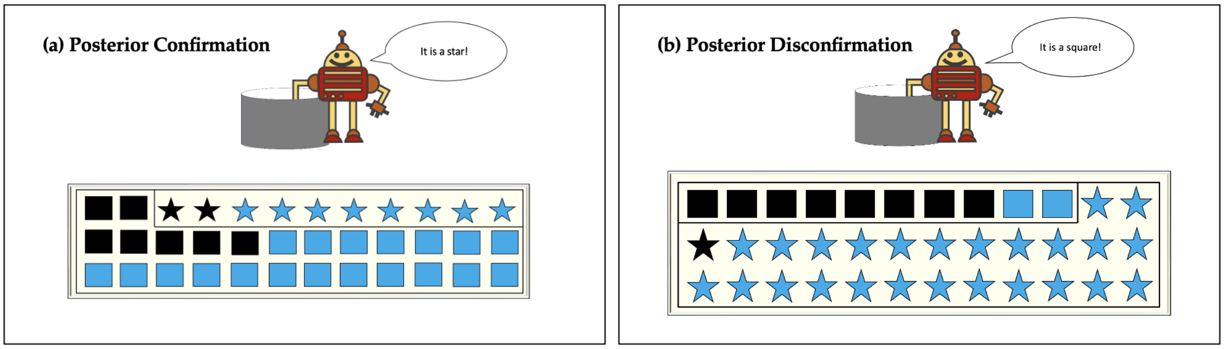

Figure 10. **Experiment 3 posterior trials.** The posterior confirmation and disconfirmation trials are equally probabilistic (8 objects vs. 2 objects). (a) In the posterior confirmation trial, the hint that the block is a star will confirm their prior probability guess (blue). (b) In a posterior disconfirmation trial, the hint that the block is a square will contradict the child's prior probability guess (children should now guess black).

Could children's success in Experiments 1 and 2 be driven by probability matching rather than the ability to reason about posterior probabilities? Probability matching would predict that children distribute their guesses in proportion to the likelihood of each outcome occurring. Specifically, children would be expected to achieve approximately 75% accuracy on prior probability trials (reflecting the 3:1 color ratio of all blocks), 100% accuracy on posterior confirmation trials (where all shapes are the same color when isolating a posterior confirmation trial), and approximately 75% accuracy on posterior disconfirmation trials (reflecting the 3:1 color ratio in those trials). To distinguish between probability matching and probabilistic reasoning, I analyzed the performance of children who answered at least three out of four prior trials correctly. Their accuracy on prior probability guesses was significantly above the predicted 75% threshold for probability matching, suggesting that their responses reflected probabilistic reasoning rather than mere probability matching.

In conclusion, my findings suggest that 4- to 6-year-olds can reason about posterior probabilities and update their decisions by integrating new information, and methods such as increasing set sizes and increasing ratios may explain the increased performance of children in my study compared to previous studies. This study provides insights into the emergence of the ability to reason about posterior probabilities and suggests that this ability may develop earlier than previously thought. Future research should explore whether removing barriers, such as small set sizes, can further enhance children's performance on such tasks and investigate whether even younger children can reason about posterior probabilities. This will contribute to a clearer understanding of the developmental trajectory of this critical cognitive skill.

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